

Tactical Dogfight Pathing

A Minimax Pursuit-Evasion Agent for 2D Grid-Based Fighter-Jet Combat

BCSE306L — Artificial Intelligence | Course Project — Review 1 Proposal

1. Problem Statement

We model a 2-D grid-based aerial dogfight as a two-player, zero-sum pursuit-evasion game: each of two fighter jets chooses a heading and moves one cell per turn, and a jet wins by manoeuvring onto its opponent's tail — within weapons range and inside its firing cone — before it is itself outmanoeuvred.

2. Module and Technique

Module 3 — Game Playing. We implement Minimax search with Alpha-Beta pruning to select each jet's move, searching several plies ahead over the grid-and-heading state space. The AI core we write ourselves is the game-tree search and a custom positional evaluation function; no library or pretrained model performs the decision-making.

3. AI Formulation

Component	Definition for this project
States	$(x_A, y_A, h_A, x_B, y_B, h_B, t)$ — grid position and heading (0 – 315° in 45° steps) of jet A and jet B, plus turn count t
Actions	{Turn Left 45° , Hold Heading, Turn Right 45° } then move one cell forward along the new heading (3 actions per ply)
Transition model	Apply the chosen turn, update heading, then advance one cell in that heading (clipped to grid bounds)
Goal / terminal test	A jet wins when the opponent is within weapons range R and inside its forward firing cone ($\text{angle-off} \leq \theta$); draw at turn limit T
Utility / evaluation	$f(s) = w_1 \cdot (\text{angle-off advantage}) + w_2 \cdot (\text{range-to-envelope score}) + w_3 \cdot (\text{edge-proximity penalty})$, computed as attacker score – defender score

4. Input and Output

Input: grid dimensions, starting position/heading for both jets, weapons range R , firing-cone angle θ , search depth, and the evaluation-function weight settings (w_1, w_2, w_3) for each agent. Output: the sequence of moves each jet makes turn-by-turn, the game outcome (win/draw), and — for the experiment — aggregate win-rate statistics across many simulated matches.

5. Experimental Comparison

We will pit agents configured with different evaluation-function weightings against each other (e.g., an angle-off-dominant heuristic vs. a range-dominant heuristic vs. a balanced one) across a batch of randomized starting configurations, and report win rate, average game length, and nodes explored per move for each configuration as a table and a bar chart, with analysis of which heuristic emphasis produces the strongest tactical play.

6. Team

Team Member	Registration No.	Role
Parth Khanayat	24BDS0299	Game engine, Minimax + Alpha-Beta core
Kushal Goel	24BCE2106	Evaluation function design & heuristic tuning
Jia Kumar	24BDS0228	Experiment harness, tournament simulation, report

7. Declaration

No external AI service (ChatGPT, Gemini, Claude, etc.) will be used to generate the reasoning component of this project; if used at all, it will be limited to a supporting role (e.g., drafting synthetic test scenarios) and will be declared in the final report.